

TAIN RANGE, United Kingdom

WMO: 030620

Lat: 57.8189N

Lon: 3.9667W

Elev: 4

StdP: 101.28

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.4	(c) -3.4	(d) -6.4	(e) 2.2	(f) -4.1	(g) -4.8	(h) 2.5	(i) -1.6	(j) 17.1	(k) 6.3	(l) 14.6	(m) 5.7	(n) 1.5	(o) 250	(p) 0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.2	(c) 21.1	(d) 16.5	(e) 19.6	(f) 15.5	(g) 18.4	(h) 14.8	(i) 17.5	(j) 20.0	(k) 16.4	(l) 18.7	(m) 15.5	(n) 17.6	(o) 5.5	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.4	(b) 11.7	(c) 18.6	(d) 15.4	(e) 10.9	(f) 17.5	(g) 14.5	(h) 10.3	(i) 16.6	(j) 49.2	(k) 20.0	(l) 46.1	(m) 18.7	(n) 43.5	(o) 17.7	(p) 28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.8	(b) 11.2	(c) 9.9	(e) -8.2	(f) 24.9	(g) 3.0	(h) 1.6	(i) -10.4	(j) 26.1	(k) -12.2	(l) 27.0	(m) -13.9	(n) 27.9	(o) -16.1	(p) 29.1		
(a) 12.8	(b) 11.2	(c) 9.9	(e) -8.4	(f) 20.3	(g) 3.1	(h) 2.7	(i) -10.6	(j) 22.3	(k) -12.4	(l) 23.9	(m) -14.1	(n) 25.4	(o) -16.4	(p) 27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	3.5	3.9	5.4	7.3	9.4	12.2	14.1	14.0	12.0	9.0	5.8	3.4
	DBStd	4.73	3.00	2.98	2.72	2.70	2.53	2.30	2.05	2.11	2.28	2.76	3.08	3.76
	HDD10.0	1044	202	171	143	88	41	6	0	1	7	52	129	206
	HDD18.3	3636	460	403	400	330	276	183	131	136	189	289	376	463
	CDD10.0	452	0	1	2	8	23	73	128	124	68	21	3	1
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	10	0	0	0	0	0	1	5	4	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
(14) Wind	WSAvg	4.7	5.0	5.3	5.3	5.0	4.6	4.5	4.0	4.0	4.3	4.7	4.7	4.7
(15) Precipitation	PrecAvg	784	76	62	56	55	62	54	62	64	56	85	74	73
	PrecMax	902	153	122	111	100	122	110	117	135	97	199	108	114
	PrecMin	636	19	22	10	22	17	21	22	21	22	27	34	20
	PrecStd	75	36	22	24	20	27	22	27	26	22	40	20	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.4	14.6	17.2	19.7	21.8	23.7	23.3	20.9	17.6	14.6	13.0
		MCWB	9.7	10.5	10.3	12.7	14.4	15.7	18.4	18.6	16.5	14.1	12.0	10.9
	2%	DB	10.6	12.0	12.5	15.1	17.5	19.4	21.2	20.8	18.7	15.8	12.9	11.5
		MCWB	8.7	9.7	9.2	11.2	13.2	14.6	16.8	16.6	14.8	13.0	10.7	9.6
	5%	DB	9.3	10.4	11.2	13.5	15.6	17.8	19.5	19.2	17.3	14.6	11.6	10.3
		MCWB	7.6	8.3	8.5	10.0	12.1	13.9	15.8	15.4	14.0	12.1	9.8	8.4
	10%	DB	8.0	8.7	10.0	12.1	14.1	16.5	18.2	17.9	16.2	13.4	10.6	8.9
		MCWB	6.5	6.8	7.5	9.2	11.0	13.1	15.0	14.7	13.3	11.3	9.0	7.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.1	11.3	13.8	15.7	17.0	19.5	19.3	17.2	14.8	12.4	11.1
		MCDB	11.5	13.0	13.8	16.1	18.2	19.8	21.8	21.7	19.5	16.7	14.2	12.6
	2%	WB	9.0	9.8	9.7	11.7	13.8	15.4	17.7	17.2	15.7	13.6	11.1	9.8
		MCDB	10.3	11.7	12.0	14.2	16.6	18.1	20.2	19.9	17.9	15.3	12.5	11.3
	5%	WB	7.7	8.5	8.7	10.5	12.6	14.5	16.6	16.2	14.7	12.6	10.0	8.6
		MCDB	9.1	10.3	10.9	13.0	15.0	17.0	18.8	18.3	16.7	14.1	11.3	10.1
	10%	WB	6.6	7.0	7.7	9.4	11.4	13.6	15.6	15.3	13.8	11.7	9.1	7.5
		MCDB	8.0	8.6	9.7	11.7	13.6	15.9	17.6	17.3	15.5	13.1	10.4	8.9
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	6.2	7.0	7.6	7.9	7.1	7.2	7.5	7.5	7.0	6.0	5.7
		MCDBR	5.9	7.3	8.7	9.7	10.4	9.6	9.9	9.4	9.1	7.9	7.0	6.7
		MCWBR	5.0	5.8	6.2	6.4	6.7	6.0	6.2	5.9	6.2	6.0	5.9	5.9
	5% WB	MCDBR	5.8	7.0	8.0	9.2	9.7	8.8	9.0	8.4	8.3	7.6	6.9	6.8
		MCWBR	5.1	5.8	6.1	6.6	6.9	6.0	6.4	5.9	6.4	6.3	6.1	6.0
(40) Clear-Sky Solar Irradiance	taub		0.314	0.337	0.359	0.379	0.379	0.375	0.378	0.366	0.353	0.338	0.315	0.301
	taud		2.215	2.306	2.314	2.303	2.352	2.396	2.398	2.439	2.473	2.422	2.269	2.152
	Ebn at Noon		520	694	784	829	855	864	851	837	793	691	517	407
	Edh at Noon		51	72	93	110	112	109	106	95	80	64	48	40
(44) All-Sky Solar Radiation	RadAvg		0.39	1.00	2.04	3.43	4.63	4.54	4.32	3.44	2.37	1.24	0.52	0.26
	RadStd		0.05	0.11	0.19	0.28	0.30	0.40	0.36	0.22	0.19	0.12	0.04	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page